MIL-E-1/1614A
12 December 1974
SUPERSEDING
MIL-E-1/1614
19 January 1966

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 857B

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Half-wave mercury rectifier

See figure 1

Mounting position: Base down

Weight: 3-1/2 pounds (1.59 kilograms)

ABSOLUTE RATINGS:

Parameter:	Ef	epx.	ib	Io	T Hg	tk	Frequency
Unit:	Vac (Note 1)	kv	a	Adc	°C	sec	Hz
Maximum:	5.25	22	40	10	40	---	150
Minimum	4.75	---	---	---	30	60	25
Maximum:	5.25	10	40	10	60	---	150
Minimum:	4.75	---	---	---	25	60	25
TEST CONDITIONS:	5.0	25	---	---	30 to 40	60	60

GENERAL:

Qualification - Required

Ⓐ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 1								
1231	Peak emission by voltage drop	-	i_b = 40 a	0.65	II	etd	---	20	v
1301	Filament current	-		0.65	II	If	27	33	Aac
1353	Operation of rectifiers	2, 3	I_o = 10 Adc (min); t = 120	0.65	II	---	---	---	---
	Quality conformance inspection, part 3								
---	Service-life guarantee	-	t = 1,000 hours	---	---	---	---	---	---

NOTES:

1. Increased life may be obtained by using filament 90° ±30° out of phase with anode excitation.
2. Artificial control of condensed mercury temperature may be used for this test, half specified inverse voltage may be applied initially and then gradually increased to full voltage in approximately 1 minute.
3. This test is to be the first test performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 17, 80
DSA - ES

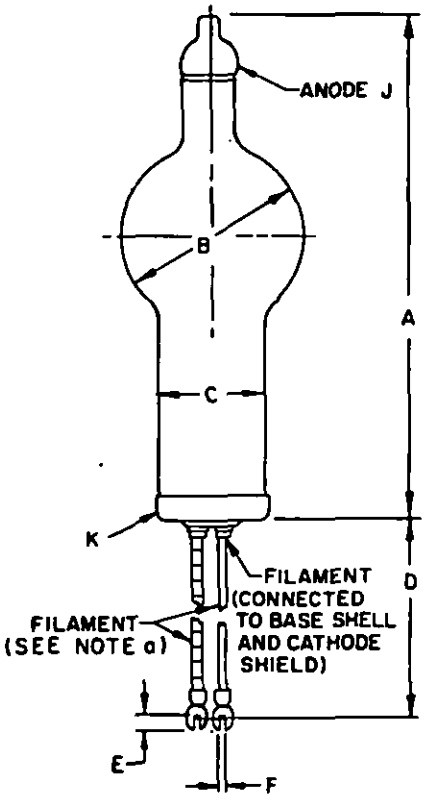
User activities:
Army - MU
Navy - AS, OS, MC, CG, SH

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2887)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	19.120 (485.65)	19.880 (504.95)
B		7.130 (181.10)
(A) Quality conformance inspection, part 3 (see note b)		
C		4.130 (104.90)
D	7.500 (190.50)	10.000 (254.00)
F	.310 (7.87)	.380 (9.65)
J	Cap: C1-10 (EIA)	
(A) K	Base: F0-2 or F0-1 (EIA)	
Reference dimension		
E	.625 (15.88)	

NOTES:
a. Material used for insulated sleeve may be beaded ceramic or vinyl coated fiberglass or equivalent.
b. A sample size of three tubes shall be checked annually, with no failures allowed. In case of a sample failure, the test shall become a quality conformance inspection (QCI), part 2 test, for three successful consecutive submissions at which time the test may revert to the QCI, part 3 test.

FIGURE 1. Outline drawing of electron tube type 857B.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

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